



PAMANTASAN NG LUNGSOD NG SAN PABLO
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Decimal to Binary



Learning Objectives

At the end of the lesson, the students must be able to:

- Explain the base-2 (binary) system and how it differs from the base-10 (decimal) system.
- Identify the steps involved in converting a decimal number to binary.
- Convert small decimal numbers (e.g., from 1 to 15) to binary by manually dividing by 2.



Decimal Number System

- it employs 10 symbols to represent numbers with a base of ten: 0,1,2,3,4,5,6,7,8, and 9.
- also known as Hindu-arabic number system.
- it is the most widely used number system, with numbers that can be easily identified even if the base is not stated.



Binary Number System

- It is base-2 number system in which numbers are only represented by two digits: 0 and 1.
- It is abbreviated as BIT which is the smallest unit of data in a computer.
- it is important to know that the Most Significant Bit (MSB) is the
bit on the far left end of a binary number, while the Least Significant Bit (LSB) is the bit on the far right end.



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Base of the
required
number

2

100

Decimal
number

Quotient

2

50

0

Remaider

2

25

0

2

12

1

2

6

0

2

3

0

1

1

$(100)_{10}$ in Binary
is $(1100100)_2$



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Decimal Numbers to Binary Numbers

- Step 1: **Divide** the given decimal number by “2” where it gives the result along with the remainder.
- Step 2: If the given **decimal number is even**, then the result will be whole and it gives the **remainder “0”**
- Step 3: If the given **decimal number is odd**, then the result is not divided properly and it gives the **remainder “1”**.
- Step 4: By placing all the remainders in order in such a way, the **Least Significant Bit (LSB) at the top** and **Most Significant Bit (MSB) at the bottom**, the required binary number will be obtained.



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Example

Let us convert the given decimal number 294 into a binary number.



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Divide by 2	Result	Remainder	Binary Value
$294 \div 2$	147	0	0 (LSB)
$147 \div 2$	73	1	1
$73 \div 2$	36	1	1
$36 \div 2$	18	0	0
$18 \div 2$	9	0	0
$9 \div 2$	4	1	1
$4 \div 2$	2	0	0
$2 \div 2$	1	0	0
$1 \div 2$	0	1	1 (MSB)



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Therefore, the binary equivalent for the given decimal number 294 is 100100110.

$$294 = 100100110$$



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Decimal to Binary Table

Decimal Number	Binary Number
0	0
1	1
2	10
3	11
4	100
5	101
6	110
7	111
8	1000
9	1001
10	1010



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Example

Convert 160 to binary number.



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Solution:

Given: Decimal Number = 160_{10}

Divide by 2	Result	Remainder	Binary Value
$160 \div 2$	80	0	0 (LSB)
$80 \div 2$	40	0	0
$40 \div 2$	20	0	0
$20 \div 2$	10	0	0
$10 \div 2$	5	0	0
$5 \div 2$	2	1	1
$2 \div 2$	1	0	0
$1 \div 2$	0	1	1 (MSB)

Therefore, $160_{10} = 10100000_2$



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Example

Convert 195.25 into binary.



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Solution:

$$195 / 2 = 97 \text{ with remainder } 1$$

$$97 / 2 = 48 \text{ with remainder } 1$$

$$48 / 2 = 24 \text{ with remainder } 0$$

$$24 / 2 = 12 \text{ with remainder } 0$$

$$12 / 2 = 6 \text{ with remainder } 0$$

$$6 / 2 = 3 \text{ with remainder } 0$$

$$3 / 2 = 1 \text{ with remainder } 1$$

$$1 / 2 = 0 \text{ with remainder } 1$$

Thus, the binary equivalent of 195 is 11000011.



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Now, we have to convert the fractional part of the given number into binary.

Multiply 0.25 by 2 and observe the resulting integer and fractional parts. Renew multiplying the resultant fractional part by 2 until we get a resulting fractional part equal to zero.

Then we need to write the integer parts from the results of each multiplication to make the equivalent binary number.

$$0.25 \times 2 = 0 + 0.5$$

$$0.5 \times 2 = 1 + 0$$

Here, 0.25 is equivalent to the binary number 0.01.

$$\text{Therefore, } (195.25)_{10} = (11000011.01)_2$$



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Decimal to Binary Practice Problems

1. Convert 244 to its equivalent binary number.
2. Convert 76 to binary number.
3. What is the binary equivalent of decimal number 891?
4. Convert 57 into a binary number.



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REFERENCE:

<https://byjus.com/maths/decimal-to-binary/>

<https://www.youtube.com/watch?v=hEZDMzWgWUM>